

Chemical Reactions

AI Tutor - Chemistry - Lesson 3

What Happens in a Reaction?

A chemical reaction occurs when substances (reactants) transform into new substances (products) by breaking and forming chemical bonds. The atoms are rearranged, but none are created or destroyed.

Signs of a Chemical Reaction

- Color change
- Formation of a gas (bubbles)
- Formation of a precipitate (solid)
- Change in temperature (releasing or absorbing heat)

Balancing Equations

Because matter is conserved, a balanced chemical equation has the same number of each type of atom on both sides. For example: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, where 4 hydrogen atoms and 2 oxygen atoms appear on both sides.